

Group Theory in a Nutshell for Physicists

Chapter 1.2 Exercises

1. Show that for 2-cycles $(1a)(1b)(1a) = (ab)$

We can write $(1a)(1b)(1a)$ as a product of exchanges as in eq (1) of ^{the} chapter:

$$\begin{pmatrix} 1 & a & b \\ a & 1 & b \\ a & b & 1 \\ 1 & b & a \end{pmatrix} = (1)(ab) = (ab)$$

$$\Rightarrow (1a)(1b)(1a) = (ab)$$

Alternatively we can just calculate using rules for multiplying 2-cycles:

$$(1a)(1b)(1a) = (1a)(b1a) = (a1)(1ab) = (1)(ab) = (ab)$$

2 Show that A_n for $n \geq 3$ is generated by 3-cycles, that is, any element can be written as a product of 3-cycles.

A_n is even permutation of n objects.

Since A_n is even it can be written as a product of even number of 2-cycles:

For A_3 we can write for example $(ab)(bc) = (abc)$ which is a 3-cycle. Since A_3 is for even permutation A_3 only has elements of

form $(ab)(bc)$ so this is true for A_3 .

For A_n we have

$$A_n: (2)(2) \cdot (2)(2) \cdot \dots \cdot (2)(2) \quad , \quad \text{where } (2) \text{ is a two cycle}$$

We can always write product of two 2-cycles as product of 3-cycles

$$\text{for example } (ab)(cd) = (acb)(adb)$$

Since A_n is made of elements that are products of even amount of 2-cycles & we can write product of two 2-cycles as product of 2 or more 3-cycles, A_n is generated by 3-cycles.

3. Show that S_n is isomorphic to a subgroup of A_{n+2} .
Write down explicitly how S_3 is a subgroup of A_5 .

S_n has even & odd permutations for n elements.
 A_{n+2} has only even permutations for $n+2$ elements.

S_n has $n!$ elements while A_{n+2} has $\frac{(n+2)!}{2}$ so A_{n+2} is bigger
both have same group multiplication.

A_{n+2} naturally has the even permutations of S_n which are A_n
since the element of A_n such as $(a\ b)(c\ d)$ is in A_{n+2} as
 $(a\ b)(c\ d)(e)(f)$.

Since A_{n+2} is permutation of $n+2$ elements we can represent
odd permutations of S_n by elements of A_{n+2} where the permutation
of the first n -elements (that are also in S_n) are odd along with
two elements (that aren't in S_n) also being odd. I.E

$(1\ 2)(3\ 4)(5\ 6)$ of S_6 can be represented by

$(1\ 2)(3\ 4)(5\ 6)(7\ 8)$ of A_8

in the subgroup of A_8 we would have these elements representing
the odd permutations of S_6 while the cycles that represent A_6 are of
the form $(1\ 2)(3\ 4)(5)(6)(7)(8) = (1\ 2)(3\ 4)$.

With some idea we can represent all permutations in S_n by
permutations in A_{n+2} . And since S_n & A_{n+2} have same group
multiplication the collection of elements of A_{n+2} representing S_n
form a Group which is of course subgroup of A_{n+2}

$$\Rightarrow S_n \cong A_{n+2} \quad \square$$

S_3 has $3! = 6$ elements & A_5 has $\frac{5!}{2} = 60$ elements

$$S_3: \{I, (1\ 2), (1\ 3), (2\ 3), (1\ 2\ 3), (1\ 3\ 2)\}$$

The elements in A_5 that form a group G_3 isomorphic to S_3 are:

$$G_3: \{I, (1\ 2)(4\ 5), (1\ 3)(4\ 5), (2\ 3)(4\ 5), (1\ 2\ 3), (1\ 3\ 2)\}$$

$$\Rightarrow S_3 \cong G_3$$

4. List the partitions of 5

$1+1+1+1+1$, $3+2$, $2+2+1$, $3+1+1$, $2+1+1+1$, 5
 $4+1$

5. Count the number of elements with a given cycle structure.

This problem description can be found in pages 59-60.

6. S_5 has $5!$ elements and the cycle structure is the following.

$$5! = 120$$

The possible cycle structures are:

Identity I

(xx) 2-cycle $\frac{5!}{2 \cdot 3!} = 10$

$(xx)(xx)$ product of 2-cycles $\frac{5!}{2 \cdot 1!} = 15$

$(xxx)(xx)$ product of 3-cycle & 2-cycle = 20 Must be same as next one for S_5

(xxx) 3-cycle = 20 (calculated below)

$(xxxx)$ 4-cycle = 30

$(xxxxx)$ 5-cycle = $5! - \sum \text{previous} = 120 - [10 + 15 + 20 + 20 + 30 + 1] = 24$

$$4! =$$

(123) and $(123)(45)$ are equivalent

(132) & $(132)(45)$

$$\begin{aligned} \binom{5}{4} \cdot (4-1)! &= \frac{5!}{4! (5-4)!} \cdot (4-1)! \\ &= \frac{5!}{4!} (4-1)! \\ &= \frac{5! \cdot 3!}{4!} = \frac{5!}{4} = 30 \end{aligned}$$

$$S_3: (12), (13), (23), \quad \frac{3!}{2} = 3$$

$$S_4: \{S_3\}, (14), (24), (34), \quad \text{Previous} + (\text{count} - 1) = 6$$

$$S_5: \{S_4\}, (15), (25), (35), (45), \quad \text{Previous} + (\text{count} - 1) = 10$$

$$\frac{4!}{4} = 6 \quad 4 \cdot 3 \cdot 2 = 12 - 2 = 2 \cdot 2 \quad = \frac{4!}{2 \cdot 2!}$$

$$\frac{5!}{12} = 6 = \frac{5!}{2 \cdot 3!}$$

$$S_6: \text{Previous} + (\text{count} - 1) = 15 \quad \frac{6!}{2 \cdot 4!} = 15 \Rightarrow \frac{n!}{2 \cdot (n-2)!}$$

Now we need to actually confirm this, we can do so via induction:

$$\frac{(n-1)!}{2 \cdot (n-3)!} + (n-1) = \frac{(n-1)!}{2 \cdot (n-3)!} + \frac{2 \cdot (n-1)(n-3)!}{2(n-3)!}$$

$$= \frac{2(n-1)(n-3)! + (n-1)!}{2 \cdot (n-3)!}$$

$$= \frac{2(n-1)(n-2)(n-3)! + (n-2)(n-1)!}{2 \cdot (n-2)(n-3)!}$$

$$= \frac{2(n-1)! + n! - 2(n-1)!}{2(n-2)!} = \frac{n!}{2(n-2)!}$$

So we get the number of elements for one 2-cycle of S_n

via formula $\frac{n!}{2(n-2)!}$ which we proved via induction.

Now for $(xx)(xx)$ we need to think more:

Start from S_4 :

$$S_4: (12)(34), (13)(24), (14)(32) = 3$$

$$S_5: \{S_4\} + (15) \overbrace{[(23), (24), (34)]}^{S_3} + (25) \cdot \{S_3\} \\ + (35) \cdot \{S_3\} + (45) \cdot \{S_3\} = \text{Previous} + (\text{current} - 1) \cdot |S_{n-2}| = 15$$

$$S_6: \{S_5\} + [(16) + (26) + (36) + (46) + (56)] \cdot \{S_4\} = 15 + 5 \cdot 6 = 45$$

$$S_7 = 45 + 6 \cdot |S_5| = 45 + 60 = 105$$

$$\text{Previous} + (\text{current} - 1) \cdot \frac{(n-2)!}{2(n-4)!} \quad \parallel \quad \begin{array}{l} \text{For } S_4: \text{ previous: } 0 \\ \Rightarrow \text{ Count right} \end{array}$$

$$\text{Previous} + (n-1) \cdot \frac{(n-2)!}{2(n-4)!} = \text{Previous} + \frac{(n-1)!}{2(n-4)!}$$

$$\frac{(n-1)!}{2(n-4)!} \stackrel{n=4}{=} \frac{3!}{2 \cdot 0!} = 3$$

$$n = 4, 5, 6, 7 \\ 3, 15, 45, 105$$

$$\frac{4!}{8} = 3, \quad \frac{5!}{8} = \frac{5!}{8 \cdot (5-4)!} = 15$$

$$\downarrow \\ \frac{4!}{8(4-4)!} = 3, \quad \frac{6!}{8(6-4)!} = \frac{6!}{8 \cdot 2} = \frac{6!}{16} = 45$$

$$\frac{6!}{8 \cdot 2} = \frac{6!}{4 \cdot 4} = 45$$



$$\boxed{\frac{n!}{8 \cdot (n-4)!}}$$

$$\frac{7!}{8 \cdot (7-4)!} = \frac{7!}{8 \cdot 3!} = 105$$

$$\begin{aligned}
& \frac{(n-1)!}{8 \cdot (n-5)!} + (n-1) \cdot \frac{(n-2)!}{2 \cdot (n-4)!} \\
&= \frac{(n-1)!(n-4) + 4(n-1)!}{8 \cdot (n-4)!} = \frac{-4(n-1)! + 4(n-1)! + n(n-1)!}{8 \cdot (n-4)!} \\
&= \frac{n!}{8 \cdot (n-4)!} \quad \square \quad \text{Proof by induction.}
\end{aligned}$$

S_5 the ones with $(x \times x)$:

$$(123), (124), (125)$$

$$S_3 : (123), (132) = 2$$

$$(132), (142), (152)$$

$$S_4 : \{S_3\}, (124), (134), (423), (142), (143), (432)$$

$$\frac{4!}{(4-1)!} = \frac{4!}{3!} = 4 = 2 + 2 \cdot 3 = 8 = \text{prev} + \left(\frac{4!}{(4-2)!} \right)$$

$$\begin{aligned}
S_5 &= \{S_4\} + [(125), (135), (145), (235), (245), (345)] \cdot 2 \\
&\quad \text{for permuting of } (a, i) \\
&= 8 + 6 \cdot 2 = 20
\end{aligned}$$

7. Show that $\mathbb{Q}$ forms a group ($\mathbb{Q}$ being the Quaternionic group)

$\mathbb{Q}$ consists of elements $1, -1, i, -i, j, -j, k, -k$ with group multiplication given by (1 is identity & $-1 \cdot (-1) = 1$)

$$i^2 = j^2 = k^2 = -1 \quad \text{and} \quad ij = -ji = k, \quad jk = -kj = i, \quad ki = -ik = j$$

From these rules we see that $g \cdot g' \in \mathbb{Q}$ for all $g, g' \in \mathbb{Q}$

Now we need to show $(g \cdot g') \cdot h = g \cdot (g' \cdot h)$

$$ijk = (ij)k = k \cdot k = k^2 = -1$$

$$ijk = i(jk) = ii = i^2 = -1$$

$$iji = i(ji) = i(-k) = -ik = j \quad \swarrow \quad ij i = (ij) i = ki = j$$

From the multiplication rule we get $abc = -1$ ^{no matter the multiplication order} for $a \neq b \neq c$ & $a, b, c \in \{i, j, k\}$
 so abc are associative

$a b a$ either $-b$ or b and associative. (Normal multiplication is associative)

$\mathbb{Q}$ has identity 1 every element has inverse eg

-1 is its own inverse & i has $-i$, j has $-j$, k has $-k$

Since elements in $\mathbb{Q}$ are associative & there exists identity & every element has its inverse, $\mathbb{Q}$ forms a group.

8. Show that A_4 is not simple

A simple group doesn't contain a (nontrivial) invariant subgroup.

Invariant subgroup H of G denoted $H \triangleleft G$ is a subgroup for which
 $g^{-1} H g = H$ for all $g \in G$.

To show A_4 is not simple we need to show it contains a nontrivial invariant subgroup

We know

$$\mathbb{Z}_2 \otimes \mathbb{Z}_2 = \{ I, (12)(34), (13)(24), (14)(23) \}$$

$$\mathbb{Z}_2 = \{ I, a \} \quad a^2 = I$$

More familiarly

$$\mathbb{Z}_2 \otimes \mathbb{Z}_2 = \{ (I, I), (I, a), (a, I), (a, a) \}$$

$$(12)(34) \cdot (13)(24) = (12)(43)(31)(42) = (14)(23)$$

So indeed the permutations $I, (12)(34), (13)(24), (14)(23)$

form a subgroup of A_4 Now we need to show that this

Subgroup is a invariant subgroup of A_4 . The group A_4 is

$$A_4: \{ I, (12)(34), (13)(24), (14)(23), (123), (132), (124), (142), (234), (243), (134), (143) \}$$

$$[(12)(34)]^2 = I, \quad (123)^{-1} = [(12)(23)]^{-1} = (13)(32) = (132)$$

$$(132)^{-1} = (123) \quad \text{etc...}$$

The elements of $Z_2 \otimes Z_2$ are $\{ I, (12)(34), (13)(24), (14)(23) \}$

Now let's show that $g^{-1}(Z_2 \otimes Z_2)g = Z_2 \otimes Z_2$ for all $g \in A_4$

$$g I g^{-1} = I \quad \text{for all } g \in A_4$$

$$\begin{aligned} (123)(12)(34)(132) &= (123)(143) = (14)(23) && \text{the elements} \\ (123)(13)(24)(132) &= (123)(234) = (12)(34) && (123) \text{ \& } (123)^{-1} = (132) \\ (123)(14)(23)(132) &= (123)(124) = (13)(24) && \text{give } Z_2 \otimes Z_2 \end{aligned}$$

$$\text{i.e. } (123) Z_2 \otimes Z_2 (123)^{-1} = Z_2 \otimes Z_2 \Rightarrow (132)(Z_2 \otimes Z_2)(132)^{-1} = Z_2 \otimes Z_2$$

Let's now proceed to next test:

$$\begin{aligned} (124)(12)(34)(142) &= (124)(134) = (13)(24) \\ (124)(13)(24)(142) &= (124)(123) = (14)(23) && \text{h } (124) \vee (142)^{-1} = \vee \\ (124)(14)(23)(142) &= (124)(243) = (12)(34) \end{aligned}$$

$$\begin{aligned} (234)(12)(34)(243) &= (234)(231) = (13)(24) \\ (234)(13)(24)(243) &= (234)(341) = (14)(23) && \Rightarrow (234) \vee (134)^{-1} = \vee \\ (234)(14)(23)(243) &= (234)(214) = (12)(34) \end{aligned}$$

And finally

$$\begin{aligned} (134)(12)(34)(143) &= (134)(321) = (14)(32) \\ (134)(13)(24)(143) &= (134)(124) = (12)(34) && \Rightarrow (134) \vee (134)^{-1} = \vee \\ (134)(14)(23)(143) &= (134)(342) = (13)(24) \end{aligned}$$

We have shown $g^{-1}Hg = H$ for $g \in A_4$ & $H = Z_2 \otimes Z_2 = \vee$

So $Z_2 \otimes Z_2$ is nontrivially invariant subgroup of $A_4 \Rightarrow A_4$ is not simple.

9. Show that A_n is invariant subgroup of S_n .

Since $A_n \in S_n$ such that A_n is even permutation of S_n

we only need to show $g^{-1} A_n g$ for odd $g \in S_n$

There are 12 such odd permutations g .

$$g^{-1} A_n g = (\text{odd permutation}) \circ (\text{even permutation}) \cdot (\text{odd permutation})$$

$$= (\text{odd permutation}) \cdot (\text{odd permutation})$$

$$= (\text{even permutation}) \quad \text{so} \quad g^{-1} A_n g \in A_n$$

for all $g \in S_n \Rightarrow A_n$ is invariant subgroup of S_n .

We will leave showing that it's maximal sometimes later the
it is intuitively understood why A_n is the largest invariant subgroup.

10. Show that the kernel of homomorphic map of a Group G into itself
is an invariant subgroup of G .

Let f be such map, then $f(g_1) f(g_2) = f(g_1 g_2)$ (from homomorphism)

kernel of f is the set of elements that are mapped to the
identity.

$$\text{So } g_1 \text{ in kernel if } f(g_1) = I$$

$$\text{From homomorphism also } f(I) = I$$

Associativity is satisfied by virtue of G being a group.

$$f(g) = I \quad f(g') = I \quad \Rightarrow \quad f(gg') = I$$

So the kernels form a closed set under multiplication i.e.

$$gg' \in K \quad \text{for all } f(g) = I, \quad f(g') = I$$

The identity exists ^{in kernel} since $f(I) = I$

Now we need inverse

$$f(g) = I \Rightarrow f(gg^{-1}) = f(I) = I$$

$$f(gg^{-1}) = f(g) f(g^{-1}) = I \quad \parallel f(g) = I \text{ since } g \text{ in kernel}$$

$$\Rightarrow f(g^{-1}) = I \Rightarrow g^{-1} \text{ in kernel also}$$

$\Rightarrow$ There exists inverse for all elements in kernel

$\Rightarrow$ The kernel K of homomorphic map of $G \rightarrow G$ is a subgroup of G . Now let's show it is invariant i.e. $gKg^{-1} = K$ for all $g \in G$.

$$\begin{aligned} f(gkg^{-1}) &= f(g) f(k) f(g^{-1}) = f(g) I f(g^{-1}) = f(gg^{-1}) \\ &= f(I) = I \quad \text{for all } g \in G \end{aligned}$$

$$\Rightarrow f(gkg^{-1}) = I \quad \text{for all } g \in G \quad \text{so } gkg^{-1} \in K \text{ for all } g \in G$$

$\Rightarrow K$ is invariant subgroup of G .

11. Calculate the derived subgroup of the dihedral group

The dihedral group is presented by

$$D_n = \langle R, r \mid R^n = I, r^2 = I, Rr = rR^{-1} \rangle$$

The derived subgroup is made by elements $\{x_1, x_2, \dots, x_k, \dots\}$ where

$$\{x_1, x_2, \dots, x_n\} \text{ are given by } x_i = \langle a, b \rangle = (ba)^{-1}(ab)$$

$$\begin{aligned} \text{For Dihedral group } \langle R^i, R^j \rangle &= (R^i R^j)^{-1} (R^j R^i) \quad \parallel R^i R^j \text{ commute} \\ &= I \end{aligned}$$

$$\langle r, R^i \rangle = (R^i r)^{-1} (r R^i) = r^{-1} (R^i)^{-1} r R^i = R^i r r R^i = R^i R^i$$

$$(R^i)^2 = R^{2i}, \quad \text{where } R^n = I \quad \text{so we can also write } R^{2i \bmod n}$$

↳ This means that the derived group has even element R^i & I so far.

$$\text{Last option is } \langle r, r \rangle = (rr)^{-1} (rr) = I \quad \text{since } r^2 = I$$

⇒ The derived subgroup only consists of I & R^i 's

R^{2i} for even n of $R^n = I$ we get only half of the elements of R_i since only the even powers survive

$$R^{(n/2+1)^2} = R^{n+2} = R^2 \quad \text{so only even other group}$$

$$(R^{n/2+2})^2 = R^{n+4} = R^4$$

$$(R^1)^2 = R^2$$

$$(R^3)^2 = R^6 \dots$$

$$\text{For } D_8: (R^4)^2 = R^4 R^4 = R^3$$

$$\text{For } D_4: (R^3)^2 = R^3 R^3 = R^2$$

$$\left. \begin{array}{l} R \rightarrow R^2 \\ R^2 \rightarrow I \\ R^3 \rightarrow R^2 \end{array} \right\} \text{Subgroup is just } \mathbb{Z}_2$$

$$\begin{array}{ll} \text{For odd } n & \begin{array}{l} R \rightarrow R^2 \\ R^2 \rightarrow R^n \\ R^3 \rightarrow R \\ R^n \rightarrow R^3 \\ R^5 \rightarrow I \end{array} & \begin{array}{l} R^3 R^3 = R^5 R \\ R^n R^n = R^5 R^3 \end{array} \end{array}$$

⇒ For even n we have $\mathbb{Z}_{n/2}$ as the derived subgroup
For odd n we have $\mathbb{Z}_n$ as the derived subgroup.

12. Given two group elements g & f , show that, while in general $fg \neq gf$, fg is equivalent to gf

i.e. show that for $gf \neq fg$, $gf = k'fgk$ for $f, g, k \in G$

$gf = k'fgk$ for some k . Now if $k = g^{-1}$ we get

$gf = (g^{-1})^{-1}fg(g^{-1}) = gf$ so the equivalency of fg with gf always holds since we can always use $k = g^{-1}$.

13. Prove that group of even order contain at least one element which is not identity but squares to the identity.

Group with even order has even amount of elements with one of them being identity that means there are

odd number of elements that are not identity. For a group to be a group every element needs to have its (unique) inverse. That means one of the elements of the remaining odd number of elements must be its own inverse & as such its square is the identity I .

$\Rightarrow$ There exists at least one element E with $E^2 = I$ & $E \neq I$ for group of even order

14. Using Cayley's theorem map V to a subgroup of S_4 .

List the permutations corresponding to each element of V . do same for $\mathbb{Z}_4$.

$$V = \mathbb{Z}_2 \otimes \mathbb{Z}_2 = \{ (I, I), (a, I), (I, a), (a, a) \}$$

$$= \{ I, A, B, C \} \quad \text{with} \quad C^2 = I, \quad AB = BA = C$$

The sub group of S_n with such structure is (lets denote by G)

$$G = \{ I, (12)(34), (13)(24), (14)(23) \}$$

$$f: I \rightarrow I$$

$$(a, 1) \rightarrow (12)(34)$$

$$\Rightarrow (a, a) \rightarrow (14)(23)$$

$$(1, b) \rightarrow (13)(24)$$